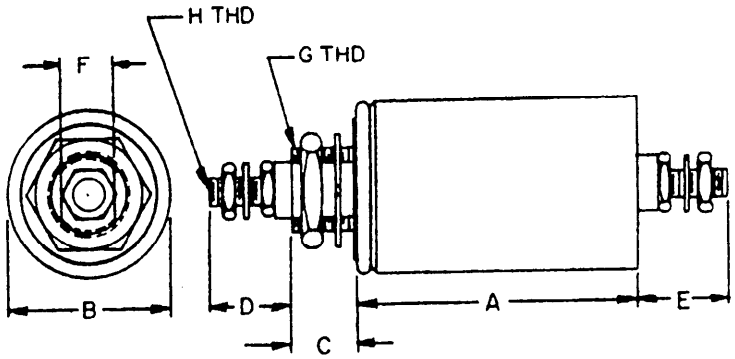
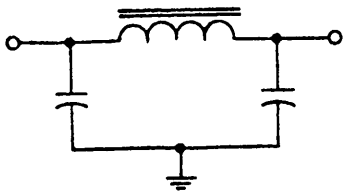
MILITARY SPECIFICATION SHEET

FILTERS, RADIO INTERFERENCE, STYLE FL86

This specification is approved for use by Rome Air Development Center, Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the filters described herein shall consist of this document and the latest issue of Specification MIL-F-15733.





TYPICAL CIRCUIT DIAGRAM

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are in parentheses.
3. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
4. Circuit diagram is for information only.
5. Mounting hardware shall be supplied with filter.

Ⓑ denotes changes.

FSC 5915

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T07:24Z
Check the source to verify that this is the current version before use.

TABLE I. Filter dimensions. (B)

Type designation 1/ 2/ 3/	Dash number	A		B		C		D	E	F		G	H
		Min	Max	Min	Max	Min	Max	Max	Max	Min	Max	Mounting thread	Terminal thread
FL86JW3JD3	0001	3.19 (81.03)	3.31 (84.07)	1.23 (31.24)	1.26 (32.00)	.480 (12.19)	.520 (13.21)	.69 (17.53)	.69 (17.53)	.520 (13.21)	.540 (13.72)	.625-24 UNC-2A	.164-32 UNC-2A
FL86JW3JD3	0002	4.94 (125.48)	5.06 (128.52)	1.73 (43.94)	1.76 (44.60)	.480 (12.19)	.520 (13.21)	.69 (17.53)	.69 (17.53)	.646 (16.41)	.666 (16.92)	.750-20 UNC-2A	.164-32 UNC-2A

1/ The letter "W" indicates insertion-loss versus frequency characteristics specified in table II.
2/ The letter "D" indicates the operating temperature range -55° to +125°C.
3/ The type designation does not describe a discrete item. For complete identification, the part number and type designation should be referenced.

TABLE II. Insertion-loss (dB) versus frequency (MHz).

Type designation	Dash number	Rated voltage	Minimum insertion loss full load in accordance with MIL-STD-220 at +25°C							
			0.15 MHz	0.3 MHz	0.6 MHz	1 MHz	10 MHz	20 MHz	40 MHz	100 MHz
FL86JW3JD3	0001 1/	100 Vdc	40	58	74	80	80	80	80	80
FL86JW3JD3	0002 1/	400 Vdc (115 Vac 0-400 Hz)	40	58	74	80	80	80	80	80

1/ Preferred for use in the design of equipment.

Source: https://assist.dla.mil -- Downloaded: 2026-01-18T07:24Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Dimensions and configuration: See figure and table I.

Case: Metal.

Terminals: Threaded stud.

Operating temperature range: -55° to +125°C; temperature characteristic D.

Rated voltage (over the rated temperature range of -55° to +125°C): See table II.

Rated current: 20 amperes dc.

Insertion loss characteristic: Symbol W (see table II).

Seal: Method 112, test condition B, MIL-STD-202.

Capacitance to ground: Method 305, MIL-STD-202. Test frequency: 1,000 ±100 Hz.

Temperature rise: 25°C, maximum.

Dielectric withstanding voltage: Method 301, MIL-STD-202. The test voltage shall be two times the rated voltage applied between either terminal and case.

Barometric pressure (reduced): Method 105, MIL-STD-202.
Test altitude: 50,000 feet.
Dielectric withstanding voltage: In accordance with initial requirements, except the test voltage shall be 125 percent of the rated dc voltage.

Insulation resistance: Method 302, test condition A, MIL-STD-202. Insulation resistance measured at 25°C between either terminal and the case shall be at least 550 megohms.

Voltage drop: 1 percent of rated voltage, maximum.

Insertion loss: In accordance with MIL-F-15733 and table II.

Overload: In accordance with MIL-F-15733.
Measurements at +25°C after test: Insulation resistance and voltage drop shall be measured and shall meet initial requirements.

Terminal strength: Method 211, test condition E, MIL-STD-202.
Applied force: 14 pounds.

Flashpoint of impregnant or potting compound: In accordance with MIL-F-15733.

Resistance to soldering heat: Method 210, test condition B, MIL-STD-202.

Salt spray (corrosion): Method 101, test condition A, MIL-STD-202.

Thermal shock and immersion: Thermal shock: Method 107, test condition A, MIL-STD-202, except that the temperature of step 3 shall be +125°C.
Immersion: Method 104, test condition A, MIL-STD-202.
Measurements after final cycling:
Dielectric withstanding voltage: As initially specified, except the test voltage shall be 90 percent of that initially specified.
Insulation resistance shall be not less than 30 percent of the initial value specified.
Insertion loss (check test) shall be measured and shall meet the initial requirements.

Source: https://assist.dla.mil -- Downloaded: 202640-18T07:24Z
Check the source to verify that this is the current version before use.

Shock (specified pulse): Method 213, test condition I, MIL-STD-202.
Mounting: Filters shall be rigidly mounted by the body.
Electrical load: 100 percent of rated voltage between either terminal and case.
Insertion loss check test: Not applicable.

Vibration, high frequency: Method 204, test condition B, MIL-STD-202. Electrical current load shall not be applied to filters during the vibration test.
Test during vibration: During the last cycle in each direction, an electrical measurement shall be made to determine intermittent-open or short-circuiting.
After the vibration test, filters shall be subjected to and shall meet the initial requirements of the insertion loss (check test).

Moisture resistance: Method 107, test condition A, MIL-STD-202, except that the temperature of step 3 shall be +125°C; no measurement shall be made before and after cycling. Filters shall then be tested in accordance with method 106, MIL-STD-202.
Initial measurements: Not applicable.
Polarization voltage: 100 Vdc.
Measurements after 24-hour drying period:
Dielectric withstanding voltage: As initially specified, except test voltage shall be 90 percent of that initially specified.
Insulation resistance shall be not less than 30 percent of the initial value specified.
Insertion-loss (check test) shall meet initial requirements.

Life: Method 108, test condition B, MIL-STD-202. Test temperature +125°C.
Test voltage: 1.4 times the rated voltage
Test current: 20 amperes.
Measurements after test:
Dielectric withstanding voltage: As initially specified, except the test voltage shall be 90 percent of that initially specified.
Insulation resistance shall be not less than 30 percent of the initial value specified.
Insertion loss (check test) shall be measured and shall meet the initial requirements.

Part number: M15733/53- (dash number from table I).

Supersession data: This specification includes the requirements of contractor (97334), specification 53B096051.

Preparing activity:
Air Force - 11

Agent:
DSA - ES

(Project 5915-F086)

Source: https://assist.dla.mil -- Downloaded: 202640-18T07:24Z
Check the source to verify that this is the currentversion before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE AIR FORCE

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 73236 WASHINGTON D. C.
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE AIR FORCE

ASD/ENESS
Wright-Patterson AFB, OH 45433

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T07:24Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 202610-18T07:24Z
Check the source to verify that this is the current version before use.